SDGP Group Project Ideas
Group 11

1)Disandu Perera

1. Emotion-Driven Music Player

- **Description: A music player that changes playlists or generates music suggestions based on the user's mood using facial expression analysis through the webcam or wearables that track heart rate.**
- **Key Features:**
  - **Real-time mood detection using AI.**
  - **Mood-based playlist generation or song suggestions.**
  - **Integration with Spotify or other music platforms.**

2. Augmented Reality Recipe Assistant

- **Description: An AR app that guides users through cooking recipes step-by-step by overlaying instructions and visuals directly on real-world objects (e.g., guiding how to chop or where to add ingredients).**
- **Key Features:**
  - **AR visualization of recipe steps.**
  - **Real-time ingredient recognition.**
  - **Interactive, hands-free navigation through the recipe process.**

3. Personal Digital Historian

- **Description: An AI-powered platform that turns your life events, conversations, and memories into a personalized digital archive. It could pull from social media, calendars, or even direct user input to build a timeline of major life moments.**
- **Key Features:**
  - **Timeline creation using various data sources (photos, texts, events).**
  - **Automatic categorization of important events and memories.**
  - **Personalized historical insights or reflection prompts based on your past.**

5. Dream Visualization and Journal Assistant

- **Description: An app that lets users describe their dreams in a simple text format and then uses AI to generate dream visualizations, helping them interpret their dreams by identifying common themes and symbols.**
- **Key Features:**
  - **Dream entry and analysis using NLP (Natural Language Processing).**
  - **AI-generated artistic visualizations of dreams.**
  - **Dream insights and analysis based on recurring themes and symbols.**

6. AI-Enhanced Interactive Storybooks

- **Description: A storytelling platform that uses AI to adapt the story based on the reader's decisions in real-time. The platform can be used for both children's books and more complex interactive novels for adults.**
- **Key Features:**
  - **Real-time branching storylines based on user interaction.**
  - **AI-driven character personalities that adapt as the story progresses.**
  - **Customizable endings based on previous choices.**

7. Personalized Nutrition Advisor with DNA Insights

- **Description: A nutrition app that provides dietary recommendations and meal plans based on a user's DNA, fitness habits, and health goals. Users can upload DNA data from existing kits (like 23andMe) for hyper-personalized advice.**
- **Key Features:**
  - **DNA-based nutrition analysis and recommendations.**
  - **Meal plans and recipes tailored to your genetic predispositions.**
  - **Integration with fitness trackers for adaptive calorie and nutrient suggestions.**

9. Crisis Simulation and Response Training App

- **Description: An app that simulates real-world crises (such as natural disasters or cyber-attacks) for individuals and businesses, offering training on how to respond effectively.**
- **Key Features:**
  - **Real-time crisis simulations based on user input and location.**
  - **Training modules for different crisis response techniques.**
  - **Post-simulation analysis with improvement recommendations.**

10. AI-Powered Home Mood Enhancer

- **Description: A smart home system that adjusts the environment based on the user's emotional state. The system would use sensors or wearables to track mood and adjust lighting, temperature, and even sound ambiance.**
- **Key Features:**
  - **Real-time emotional state detection using wearables or voice analysis.**
  - **Automated adjustment of smart home devices (lights, music, temperature).**
  - **Customizable mood settings (calming, energizing, etc.).**

11. Eco-Footprint Travel Advisor

- **Description: An app that helps users plan trips with the lowest possible carbon footprint, offering suggestions for eco-friendly accommodations, transportation, and activities.**
- **Key Features:**
  - **Travel planning tools with carbon footprint calculation.**
  - **Suggestions for eco-friendly alternatives for accommodations and activities.**
  - **Integration with carbon offset programs to reduce travel impact.**

12. AI-Based Argument Coach

- **Description: A communication training tool that helps users improve their argumentation and persuasion skills by simulating debates or conversations with an AI that adapts to different argumentative styles.**
- **Key Features:**
  - **Real-time argument simulation with AI-driven conversation partners.**
  - **Feedback on logic, tone, and persuasion effectiveness.**
  - **Tips for improving argumentation and communication strategies.**

13. Mindfulness VR Adventure

- **Description: A virtual reality game that integrates mindfulness exercises into an adventure setting. Users can explore beautiful, immersive landscapes while being guided through meditation or breathing exercises.**
- **Key Features:**
  - **VR-based interactive environments designed to promote mindfulness.**
  - **Integrated guided meditation, relaxation, and breathing exercises.**
  - **Personalized mindfulness goals and progress tracking.**

14. Remote Collaboration Fitness Challenges

- **Description: An app where teams of remote workers or friends can join fitness challenges together, track their progress, and earn rewards for staying active, fostering both health and collaboration.**
- **Key Features:**
  - **Team-based fitness challenges with real-time progress tracking.**
  - **Integration with fitness apps and wearables.**
  - **Leaderboards and rewards system for achieving group goals.**

15. Dynamic City Experience Explorer

- **Description: An app that helps travelers explore cities by dynamically generating routes and experiences based on their interests, weather, and time of day. It adjusts in real-time to provide the best possible experience.**

- **Key Features:**
    - **Dynamic route generation based on real-time data (weather, traffic).**
    - **Personalized recommendations for attractions, restaurants, and events.**
    - **Interactive maps that adjust based on preferences (e.g., art, food, history).**

2) J.A.D.D.Y Jayasuriya

IDEA 01 - Bus availability tracking system.

PROBLEM - Searching for available buses, search for bus routes, search for how much people going in particular bus as a percentage, track the expenses for particular places and trips.

Solution - This particular application can be useful for the people who use buses for day to day trips. This can be really helpful for the passengers because most of the passengers goes to their workplaces, special places and etc by those buses. So as an example let's assume that a particular person need to know what is the most suitable route for the particular place. By using this application that person can know what is the most suitable bus routes as well as the most suitable and available bus. Not only that that person can easily find out the expenses for each and every trip and the places. Also for an additional feature the user can access how many people are in that particular bus by as a percentage.

Features to implement - User's can know there current location by using the application as well as information about the current bus such as (license plate, vehicle registration number, vehicle information etc.) and it's driver.

Project overview - User's can simply know what are the specific bus routes for each specific place as well as the expenses. User can simply know his current location as well as the

IDEA 02 - Suggesting a song to the current mood.

PROBLEM - Finding the ideal music to fit there present mood can be difficult for many people. Although the right song may elevate the feelings, choosing the right one frequently. Requires thought and work. Users might not be aware of then tunes that might not always be aware of the tunes that might uplift them or provide them comfort when they're feeling down.

SOLUTION - Based on the user's current emotional state as shown by their facial expressions like frowns, grins, or neutral expressions will be detected by the system which will then suggest a playlist based on the user's emotional state. This simplifies process of choosing the appropriate music by doing away with the necessity for manual song selection. Time saving and improving the listening experience, technology makes personalized song with the recommendation.

Features to implement - playlist recommendation based on emotions: identify different emotions (such as joyful, sad or neutral) and offer in then moment music selection. Smooth integration for immediate playlist access, the system ought to be integrated with well known using music services like apple music as spotify.

Project overview - **Users are able to select the desired emotional tone(happy, suspenseful, sad) using emotion based filtering. Recommendation that are in line with the reader's mood based on sentiment analysis of the books content.**

Existing products to compare - **Spotifies daily mix and mood's playlist.**

Technology stack suggestion - **Machine learning (emotional detection)**

IDEA 03 - Suggesting a haircut

PROBLEM - **Nowadays people are busy with too much work that won't be able to know which haircut will be suitable for them.**

SOLUTION - **By using this application, user's face can be detected by the biometrics of this application and then detects which face type that user's has. Then it automatically implements the user about his/her face type. Then suggests what's the most suitable haircut for him/her.**

IDEA 04 - Suggesting a working plan

PROBLEM - **Nowadays people are so busy with work. They can have multiple amount of plans for a single day. So the people might not have enough time to make their own plans.**

SOLUTION - **By using this application user can submit many plans as they wants and then by this application, it will automatically creates a schedule of plans.**

IDEA 05 - Anxiety releasing AI companion

PROBLEM - Nowadays a lot of people are suffocating because of depression and anxiety due to different reasons.

SOLUTION - By using this AI companion it can listen to the user's problems and suggest proper solutions for it or try to make the user's current mood by complementing him/her as well as plays some calm music to keep user safe, happy and relaxed.

3)W.K.D. Bihandu Methsilu Wijesiri

01. A system of outfit recommendations.

I. Data Privacy and Ethical Use of Personal Information.

Problem: Outfit recommendation systems often rely on data such as personal preferences, body measurements, and fashion styles. Storing this information could pose privacy risks, especially if it's misused, sold to third parties, or hacked

Solution: Implement privacy-preserving techniques like anonymization and encryption of personal data. Allow users full control over what data they share, and ensure strict adherence to data protection laws. Offering a decentralized system where users store their personal information locally could also reduce risks associated with centralization.

II. Limited Fashion Inclusivity and Diversity.

Problem: A recommendation system may become biased towards popular fashion trends or mainstream brands, neglecting niche, cultural, or body-diverse fashion. This could result in recommendations that don't reflect the personal style or body type of a wide range of users.

Solution: Integrate more diverse datasets, including regional styles, sustainable fashion, and inclusive sizing options. Develop algorithms that recognize different cultural fashion trends and body shapes to offer truly personalized recommendations. Collaborating with smaller, independent designers could help diversify options.

III. Algorithm Bias and Lack of Personalization.

Problem: A recommendation algorithm might not be able to fully understand a user's unique style preference, resulting in suggestions that feel impersonal or repetitive. There may be bias in the system , especially if it favors certain body types, genders or demographics based on the dataset it was trained on,

Solution: Regularly update the recommendation algorithm to factor individual feedback and evolving preferences. Allow users to manually customize the recommendations by giving them more control over filters. Continuous user feedback should be integrated to ensure the system adapts to their evolving tastes.

02. Tourist Attraction Recommendation System.

I. Overtourism and Environmental Impact.

Problem: Overtourism can cause overcrowding at popular destinations, leading to damage to cultural landmarks, environmental degradation, and a reduced experience for visitors. A recommendation system might

unintentionally exacerbate this by pushing people toward the same popular sites repeatedly.

Solution: The system could include sustainable tourism features, recommending lesser-know of off-peak attractions to spread out visitor numbers. By incorporating environmental impact data into its algorithms, it can prioritize recommendations that support eco-friendly travel and guide users toward responsible tourism practices, such as suggesting local experiences or sustainable accommodation.

II. Bias Towards Commercialized or Popular Attraction.

Problem: Algorithms might favor the most popular or highly reviewed tourist destinations, ignoring smaller, less commercialized sites or culturally significant but lesser-known places. This could lead to homogenized travel experiences and lack of diversity in recommendations.

Solution: To combat this, the system can be designed to balance commercialized and off the beaten path recommendations. By integrating local insights and user preferences, it can offer more diverse suggestions. Allowing for user generated content, where locals can recommend their favorite spots, could also enhance authenticity.

III. Lack of integration with Local Events and Festivals.

Problem: The system may not account for temporary or seasonal events such as local festivals, parades, or cultural celebrations, which could enhance a tourist's experience but often go unnoticed in standard tourist recommendations.

Solution: Create partnerships with local tourism boards to regularly update information on events, festivals, and seasonal activities. The system could then recommend attractions based on the travel dates and align with ongoing events, providing users with a richer, time-sensitive experience.

03. Waste Management System.

I. Increased Waste Volume and Limited Landfill Space.

Problem: As global populations grow and consumption increases, the volume of waste produced is escalating. Traditional waste management systems often rely on landfills, which are finite in capacity. This can lead to an urgent crisis where available landfill space runs out.

Solution: Focus on waste reduction strategies and circular economy principles. Encourage better resource management through policies that prioritize reduction, reuse, and recycling over disposal. Implement large-scale composting programs for organic waste and adopt waste-to-energy technologies like incineration or biogas production, which can reduce landfill dependency.

II. Lack of Proper Segregation at Source.

Problem: One of the biggest challenges in waste management is ensuring that waste is properly segregated (organic, recyclable, hazardous) at the point of disposal. Without adequate separation, recycling becomes inefficient, and hazardous materials can pose environmental risks.

Solution: Strengthen public education and awareness campaigns to teach citizens the importance of waste segregation. Municipalities could also introduce stricter regulations and offer incentives, such as reduced fees or rewards for households that properly separate their waste. Additionally, creating clear labeling on products about recyclability can help guide consumers.

04. Emotion Recognition System For Psychiatrist Hospitals.

I. Inaccuracy in Emotion Recognition.

Problem: Current emotion identification algorithms often employ physiological indicators (e.g., body language or heat rate) or information from speech patterns or facial expressions to determine emotions. Without IOT devices that record data in real-time, the system could not gather enough precise data to interpret emotions in an appropriate manner, which might lead to inaccurate diagnoses or inadequate treatment plans.

Solution: Improve the system's data analysis algorithms and rely on non-invasive techniques like machine learning and audio or video analysis to boost the system's accuracy. Regularly upgrading the library of emotional expressions and incorporating feedback from psychiatrists may help enhance the system's ability to recognize emotions. Technology may be used with manual reviews to ensure correctness.

II. Context Misinterpretation.

Problem: Algorithms that recognize emotions may struggle to pinpoint a patient's emotional history, including prior encounters, significant life events, or mental health conditions like sadness or anxiety. Without the proper context, the system may misinterpret emotions or fail to recognize the complexity of mental diseases.

Solution: Include contextual data input mechanisms, such as patient history, environmental factors, or therapist notes, into the system. This could help the system understand the broader context surrounding emotional expressions. Psychiatric professionals could manually input observations or use integrated patient reports to provide contextual information.

III. Over-reliance on Technology.

Problem: Psychiatrists and hospital employees run the risk of relying too much on the system to diagnose or treat patients, which might compromise

the value of human judgment, empathy, and interpersonal communication in psychiatry.

Solution: Emotion recognition systems should be presented as helpful aids rather than as definitive diagnosis instruments. Continuous training is necessary to remind healthcare professionals that technology should complement their talents rather than replace them. The final decision will be made by human professionals, even though the technology can provide psychiatrists with real-time advice recommendations.

05. Tuition Center Management System.

I. Student Mental Health and Well-being.

Problem: Due to rising academic pressure, tuition centers may find it challenging to assist students' mental health and welfare. An academic performance-only system may cause anxiety and exhaustion.

Solution: Implementing well-being programs that offer mental health support, such regular seminars or therapy sessions, is the solution. Monitoring students' stress levels and creating a balanced workload can help prevent burnout. By fostering a happy learning environment, tuition centers may improve students' academic performance and well-being.

4)Vidura Baddage

Crowdsourced Travel Guide App

- Description: An app where travelers can share tips, reviews, and experiences about different locations around the world.
- Features:
    - User-generated travel tips and itineraries
    - Map-based navigation to nearby attractions
    - Rating system for places, restaurants, and activities
    - Option to save favorite locations and download for offline use

Online Debate Platform

- Description: Build a platform where users can debate topics in structured formats, with features for audience voting and scoring.
- Features:
    - Topic categories and debate scheduling
    - Structured debate formats (e.g., timed arguments, rebuttals)
    - Audience participation through live voting and comments
    - User rankings based on debate scores and feedback

Real-Time Emergency Vehicle Routing System

- Problem: Emergency vehicles, like ambulances and fire trucks, often face delays due to traffic congestion or roadblocks.
- Solution: Develop a real-time routing system that monitors traffic conditions and provides optimal routes for emergency vehicles. The system can integrate with traffic management systems to prioritize emergency vehicles, reducing response times.

Car Sharing Platform

- Description: A car-sharing marketplace where users can rent out their vehicles to others, providing an alternative source of income for vehicle owners.
- Economic Impact: Maximizes the utility of underused assets (cars) and reduces the need for car ownership, especially in urban areas.
- Features:
    - Car listing and booking system with availability calendars
    - Identity verification and insurance integration for safety
    - Payment processing and automatic payouts to car owners
    - GPS tracking and car condition reporting for rentals

Public Transportation Crowding Tracker

- Problem: Overcrowding in public transportation, especially in urban areas.
- Solution: Design an app that uses crowd-sourced data to monitor public transportation occupancy, helping users choose less crowded times or alternative routes.

Emergency Medical Information App

- Problem: In emergencies, critical medical information (e.g., allergies, conditions, medications) may not be readily available to first responders.
- Solution: Develop an app where users can store essential medical information that can be accessed quickly by emergency responders through a QR code or via NFC on their phones.

5)Vihari Ambegoda

Internship Finder Platform:

Description:
An Internship Finder Platform is a web-based platform designed to help students and job seekers find internship opportunities that match their skills, location, and interests. The platform connects employers and candidates by providing a streamlined process for posting

internships and applying for them. It also offers features like profile building, resume creation, and personalized recommendations based on the user's career goals.

Key Points:

1. User Profiles:
    - Allows users to create detailed profiles with their skills, education, and career interests.
    - Option to upload resumes and portfolios.
2. Search & Filters:
    - Internship listings are searchable by location, industry, duration, and required skills.
    - Personalized internship recommendations based on user preferences and skills.
3. Employer Dashboard:
    - Companies can post internship opportunities, review applicants, and manage job postings through a dedicated employer dashboard.
4. Application Tracking:
    - Users can track the status of their applications and receive updates from employers.
    - Notifications for new internships and deadlines.
5. Resume Builder & Resources:
    - Integrated tools to help users build professional resumes and cover letters.
    - Access to career resources like interview tips, skill assessments, and workshops.
6. Networking Opportunities:

    - Connect with industry professionals, mentors, and other job seekers.
    - Virtual events like career fairs, webinars, and employer Q&A sessions.
7. Mobile Accessibility:
    - Fully responsive design, allowing users to search for internships and apply directly from their mobile devices.

Student Time-Tracking App:

Description:
A Student Time-Tracking App is designed to help students manage their time efficiently by tracking how much time they spend on various academic tasks, extracurricular activities, and personal commitments. The app offers features to set study goals, schedule tasks, monitor progress, and analyze time management patterns, ultimately improving productivity and balance between schoolwork and other activities.

Key Points:

1. Task Scheduler:
    - Allows students to create and schedule tasks for assignments, study sessions, exams, and personal activities.
    - Calendar integration with reminders for upcoming deadlines and events.
2. Time Logging:

    - Users can log the time they spend on different tasks manually or with a built-in timer.
    - Categorize tasks (e.g., studying, project work, exercise) to analyze where time is spent.
3. Goal Setting:
    - Set daily, weekly, or monthly study goals for subjects or projects.
    - Track progress toward goals and get notifications when goals are met or missed.
4. Analytics & Reports:
    - Visual reports showing time spent on different activities (e.g., pie charts, bar graphs).
    - Insights into time management patterns, including areas where students are most productive or need improvement.
5. Focus Mode:
    - A distraction-free mode that blocks non-academic apps during study sessions.
    - Pomodoro timers to encourage focused work with short breaks in between.
6. Integration with Learning Platforms:
    - Sync with online learning platforms (e.g., Google Classroom, Blackboard) to automatically import assignments and deadlines.
7. Habit Building & Rewards:
    - Track habits like consistent study hours or regular breaks.
    - Earn rewards or badges for maintaining productivity streaks and achieving time-management goals.
8. Cross-Platform Access:
    - Available on mobile and desktop, allowing students to track their time across devices.

- Sync data across devices for seamless use in different environments (e.g., home, library, school).

Voting System for Overseas Citizens:

Description:
A Voting System for Overseas Citizens is a safe and easy-to-use platform that allows people living abroad to vote in their home country's elections without needing to travel back. It uses secure technologies to make sure their votes are protected, counted, and trusted.

Key Points:

1. **Secure Login:**
   - **Voters use government IDs, facial recognition, and other ways to prove who they are before voting.**
2. **Online Ballots:**
   - **Citizens can access their voting ballots online through a secure website or app, based on their home district.**
3. **Encrypted Voting:**
   - **Votes are encrypted (locked with special codes) so no one can see or change them.**
4. **Adjustable Time Zones:**

- **Voting times are set to fit different time zones, so citizens can vote from anywhere in the world.**

5. **Vote Confirmation:**
    - **Voters get a confirmation code or receipt to check that their vote was received and counted.**
6. **Fraud Protection:**
    - **The system has built-in protections to detect fake votes or suspicious activity.**
7. **Mobile & Web Access:**
    - **Voters can cast their ballots through their phones or computers, with offline options for areas with no internet.**
8. **Multiple Languages:**
    - **The system supports different languages so people from all over the world can use it easily.**
9. **Election Monitoring:**
    - **The system lets election officials track the voting process to ensure fairness and security.**
10. **Help and Support:**
    - **24/7 support is available for any voter who needs assistance, with options for people with disabilities.**

Mental Health Support App:

Description:

A Mental Health Support App is a platform designed to provide individuals with easy access to mental health resources, tools, and support. It offers features like mood tracking, guided meditation, access to mental health professionals, and a safe space for users to share their feelings. The app aims to promote

mental well-being, reduce stress, and provide timely support to those in need.

Key Points:

1. **Mood Tracking:**
    - **Users can track their emotions daily, allowing them to identify patterns and triggers over time.**
    - **Mood analytics help users understand their mental state better.**
2. **Self-Help Tools:**
    - **Guided meditations, breathing exercises, and mindfulness practices to help reduce stress and anxiety.**
    - **Journaling options to encourage self-reflection and emotional expression.**
3. **Access to Professionals:**
    - **Connect with licensed therapists or counselors for virtual therapy sessions.**
    - **Option to chat or schedule video calls with mental health professionals.**
4. **Supportive Community:**
    - **A safe, anonymous space for users to share their feelings and experiences with a supportive online community.**
    - **Moderated forums to ensure a positive, non-judgmental environment.**
5. **Crisis Support:**
    - **Immediate access to emergency contacts, helplines, or on-call counselors for users experiencing a mental health crisis.**
6. **Personalized Content:**

    - **The app suggests articles, podcasts, and videos based on users' mental health needs and preferences.**
    - **Daily affirmations or motivational messages to boost mood and confidence.**
7. **Reminders & Notifications:**
    - **Regular reminders to check in with your emotions, complete exercises, or attend therapy appointments.**
    - **Push notifications for daily self-care routines.**
8. **Data Privacy & Confidentiality:**
    - **All data is encrypted and kept confidential to ensure users feel safe and secure sharing their mental health journey.**
9. **Progress Reports:**
    - **Users can see reports on their mental health progress over time, helping them monitor improvements or areas that need more focus.**
10. **Offline Access:**

- **Some features, like meditation and journaling, are available offline so users can access support even without an internet connection.**

Personal Health Monitoring App

Problem: People struggle to keep track of their health metrics and habits.

Solution:

- **A mobile app that monitors daily activities, such as steps, sleep, water intake, and heart rate, using smartphone sensors and wearables.**
- **It provides personalized health tips, tracks progress, and gives reminders for healthier habits.**

Key Features:

- **Integration with wearables like Fitbit, Apple Watch, etc.**
- **Data visualizations (graphs and reports) for fitness progress.**
- **Custom alerts for water intake, sleep, exercise, etc.**
- **AI-based suggestions for improvements in daily routine.**

Tech Stack:

- **Frontend: React Native for cross-platform mobile apps**
- **Backend: Node.js or Flask**
- **Database: MongoDB or Firebase**

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2. Eco-Friendly Shopping Guide

Problem: Consumers struggle to identify sustainable, eco-friendly products while shopping.

Solution:

- **A mobile app or browser extension that scans product barcodes or searches for products and provides sustainability ratings, eco-friendly certifications, and environmental impact details.**

Key Features:

- **Product barcode scanner.**

- **Sustainability rating system based on company practices, material sourcing, and environmental certifications.**
- **Personalized recommendations for eco-friendly alternatives.**
- **Integration with e-commerce platforms for easy shopping.**

Tech Stack:

- **Frontend: Flutter/React Native for the mobile app or Chrome extension**
- **Backend: Node.js or Django for sustainability data**
- **Database: PostgreSQL or Firebase**